AgriCast: Crop Supply Forecasting and Allocation System for Boac, Marinduque

Sheila Mae M. Romasanta, Aron M. Zulueta, John Lloyd Q. Laylay, John Cedrick R. Malco, Benedict O. Medallon, and Ken Gabriel LB. Montellano

BSI/T - 3C | Marinduque State University

ABSTRACT

AgriCast was proposed as a crop supply forecasting and allocation system for the Office of the Municipal Agriculturist (OMA) of Boac, Marinduque. A preliminary survey conducted by the researchers identified problems in collecting updated crop information, combining barangay records, estimating harvests, and finding possible markets for available produce. Reported methods included field visits, farmer interviews, spreadsheets, paper records, and digital tools. These findings showed a need to improve how crop information was collected, checked, and used. The proposed system connects these activities through a Field Mobile App and a web dashboard. The mobile application is designed to let field personnel record planting information, view estimated harvest dates and quantities, and save records without an internet connection. The web dashboard is designed to let authorized personnel review records, monitor crop supply, compare expected harvests with estimated consumption, receive harvest reminders, and review suggested destinations for surplus produce. The proposed web interface uses HTML, CSS, JavaScript, SQL, Flutter and Leaflet. Harvest estimates are based on crop benchmarks and do not cover all factors that may affect actual production. The researchers developed the system in stages and conducted functional, integration, and unit testing. They also evaluated the system's accuracy, ease of use, and practical benefits.

Keywords: Agriculture and Food Innovation, Web-Based Crop Supply Forecasting and Allocation System, HTML, CSS, JavaScript, Flutter, and Leaflet, Crop Supply Monitoring and Surplus Allocation.

I. INTRODUCTION

1.1 Background and Problem Context.

Updated crop information helps agriculture offices monitor planting activities, estimate harvests, and prepare for possible changes in supply. Philippine Statistics Authority (PSA) reports provide information about crop production in Marinduque and MIMAROPA. These reports cover cereals, fruits, vegetables, root crops, and other crops [1],[3], [4], [5].

Agricultural information is already collected through existing government activities. PSA conducts crop-production surveys [2], while the Provincial Government of Marinduque publishes reports about its programs and activities [6]. These sources provide background information for the study. However, the researchers still need to check the tools and processes used by the OMA Boac to identify which tasks AgriCast can support or improve.

To do this, the researchers conducted a preliminary survey on how the office collected, recorded, and used crop information. Reported methods included farm visits, farmer interviews, field reports, and coordination with barangay officials. Records were kept using paper forms, logbooks, spreadsheets, and digital tools like Microsoft Excel.

The survey identified problems in obtaining updated information, estimating future harvests, combining barangay records, and identifying areas with extra or insufficient supply. It also identified difficulties in finding possible markets for available produce.

These concerns affect agriculture personnel, field workers, farmers, farmer groups, barangay officials, and other offices involved in crop monitoring and supply coordination. Existing responses include contacting farmers, coordinating with cooperatives, looking for alternative markets, and sourcing produce when supply is insufficient.

Research on digital agriculture in the Philippines discusses how technology can support data sharing, agricultural services, and decision-making. It also identifies challenges such as limited internet access and differences in users' digital skills [7]. Other studies and reports emphasize that agricultural technology should match users' needs, available resources, and working conditions [8],[9], [10], [11]. Research on digital adoption also supports providing suitable training and considering users' ability to use the technology [12], [13].

Harvest forecasting must be explained carefully. Crop yields can be affected by weather, farm management, pests, and other conditions. Studies on yield prediction use different methods and information to account for these factors [14], [15], [16]. AgriCast will use planting records and crop benchmarks to provide early estimates of harvest dates and quantities. These estimates will support crop supply monitoring and help agriculture personnel prepare for upcoming harvests and identify possible destinations for surplus produce.

Effective supply planning involves more than estimating the next harvest. It also requires reviewing production, consumption, available supply, and market conditions to guide decisions about where produce may be needed [17]. Reports on food and agriculture further highlight the importance of considering food systems, production challenges, and food prices in agricultural planning [18], [19], [20]. These studies provide a basis for AgriCast's focus on supply monitoring and coordination.

1.2 Problem Statement and Research Gap

The preliminary survey identified problems in collecting, updating, checking, and combining crop information. The office uses different recording methods and digital tools, but still faces difficulties in keeping records updated, complete, and organized.

Improving agricultural information management requires more than introducing a computer system. It also involves connecting existing data and digital tools so that information can better support farm planning and market coordination. Briones et al. emphasized the need to harmonize agricultural data, connect digital services, and strengthen decision-support tools in the Philippines [7].

AgriCast aims to support these activities through a Field Mobile App and a web dashboard. The mobile application will support field data collection, while the dashboard will support record review, harvest monitoring, and supply coordination.

The study will answer the following questions:

1. How can a Field Mobile App be developed to record planting information, provide benchmark-based harvest estimates, and support offline saving and synchronization?
2. How can a web dashboard be developed to support record review, barangay-level crop monitoring, supply comparison, harvest reminders, and suggested destinations for surplus produce?
3. Does the developed system perform its required functions and handle data correctly?
4. How close are the benchmark-based harvest estimates to the recorded actual harvests?

1.3 Objectives

General Objective

To design, develop, and evaluate a crop supply forecasting and allocation system that addresses the confirmed information-management needs of the Office of the Municipal Agriculturist of Boac, Marinduque.

The study aims to:

1. Develop a Field Mobile App that allows authorized field personnel to:
    - Record planting information.
    - View estimated harvest dates and quantities based on crop benchmarks.
    - Save records offline and upload them when an internet connection becomes available.
2. Develop a web dashboard that allows authorized agriculture personnel to:
    - Review, approve, or return submitted records.
    - View approved crop information through a barangay map with filters.
    - Compare expected harvests with estimated consumption.
    - Receive harvest reminders.
    - Review suggested destinations for surplus produce.
3. Test whether the system performs its required functions and handles data correctly, including offline saving, synchronization, calculations, record approval, and user permissions.
4. Compare benchmark-based harvest estimates with recorded actual harvests to assess their accuracy.

1.4 Proposed Solution, Scope and Limitations

AgriCast will consist of two main parts: a Field Mobile App and a Provincial Central Web Dashboard. Boac, Marinduque will serve as the pilot area.

The Field Mobile App will allow Agricultural Technicians and Agricultural Extension Workers to record planting information, view estimated harvest dates and quantities, and save records without an internet connection. Saved records will be uploaded when a connection becomes available.

The Provincial Central Web Dashboard will allow authorized municipal and provincial agriculture personnel to review, approve, or return submitted records for correction. Only approved records will be included in official summaries. Upload status and approval status will be tracked separately.

The dashboard will display approved crop information through a barangay map. Users will be able to select a barangay to view planted areas, expected harvest quantities, estimated harvest periods, estimated weekly market capacity, and projected surplus. Configurable harvest reminders will notify authorized personnel two to three weeks before the expected harvest to support advance planning and coordination.

The system will suggest how surplus produce can be allocated to possible receiving municipalities based on projected surplus, distance, and available market information. It will identify suggested destinations and quantities for review by authorized agriculture personnel.

The study will cover planting records from Boac and crops with configured maturity and yield benchmarks. It will not include transport booking, vehicle tracking, online payments, product purchasing, automated farming, or nationwide deployment.

Harvest estimates will use planting records and crop benchmarks to support supply monitoring and planning. The initial method will not directly include weather, pests, diseases, crop damage, or changes in farm management. Crops harvested more than once will require harvest schedules that distribute the expected yield across the appropriate periods.

Weekly market capacity estimates will be based on population and available per-capita crop consumption data. They will serve as planning references rather than confirmed market demand. Likewise, a suggested destination will not mean that buyers have agreed to purchase or receive the produce.

II. METHODOLOGY

2.1 Development Approach

The researchers used an incremental development approach to build AgriCast. The system was developed one feature at a time, with each feature completed before work moved to the next. Additional features were connected to the existing functions as development progressed. Changes were made to affected features when revisions were needed.

The researchers conducted a preliminary survey and prepared the initial project documents. The first stage focused on requirements checking. The researchers reviewed the survey findings, user roles, required data, and expected system behavior.

The next stage covered interface design, database design, calculation rules, and the process for uploading offline records.

Development began with the basic workflow:

1. Enter a planting record.
2. Save the record on the mobile device.
3. Upload the record.
4. Review and approve or return the record.
5. Display approved information on the dashboard.

Other features were added after the basic workflow was working. These included harvest estimates, maps, supply comparisons, reminders, and allocation suggestions.

Testing included unit, functional, and integration tests. Unit testing checked individual functions and calculations. Functional testing checked whether the system features worked according to the intended workflow. Integration

testing checked whether the connected system components worked together. Flutter documentation describes unit, widget, and integration testing, which guided the mobile testing plan .

2.2 Requirements and Core Features

Table 1. requirements and core features

Problem / need	Core feature	Acceptance criterion
Missing planting information	Standard planting form	The system rejects missing required fields and invalid values.
Unclear farm location	Barangay and coordinate recording	The correct location is saved with each record.
Limited internet access	Offline saving	Records remain available after the app is restarted without the internet.
Uploading saved records	Synchronization queue	Records upload without duplication, and failed uploads are shown.
Harvest estimation	Benchmark-based calculations	Results match independently checked calculations.
Checking submitted information	Approval process	Only approved records appear in official summaries.
Barangay monitoring	Map and filters	Displayed totals match the selected approved records.
Supply review	Harvest and consumption comparison	The displayed result matches the calculated difference.
Advance coordination	Harvest reminders	Reminders appear within the configured period.
Finding possible destinations	Allocation suggestions	Suggested quantities do not exceed the available projected surplus.
Protecting records	User permissions	Users cannot perform actions outside their assigned role.

2.3 Interface Architecture and Data Handling

AgriCast used a mobile app, web dashboard, backend API, and central MySQL database. The mobile app allowed agricultural technicians to record and upload planting information, while the web dashboard allowed authorized personnel to review, approve, and monitor records.

The Field Mobile App collected barangay, farm location/coordinates, crop, planted area, planting date, and planting method.

The Provincial Central Web Dashboard displayed approved records, barangay summaries, harvest estimates, supply comparisons, alerts, and allocation suggestions. It also allowed authorized personnel to approve or return submitted records.

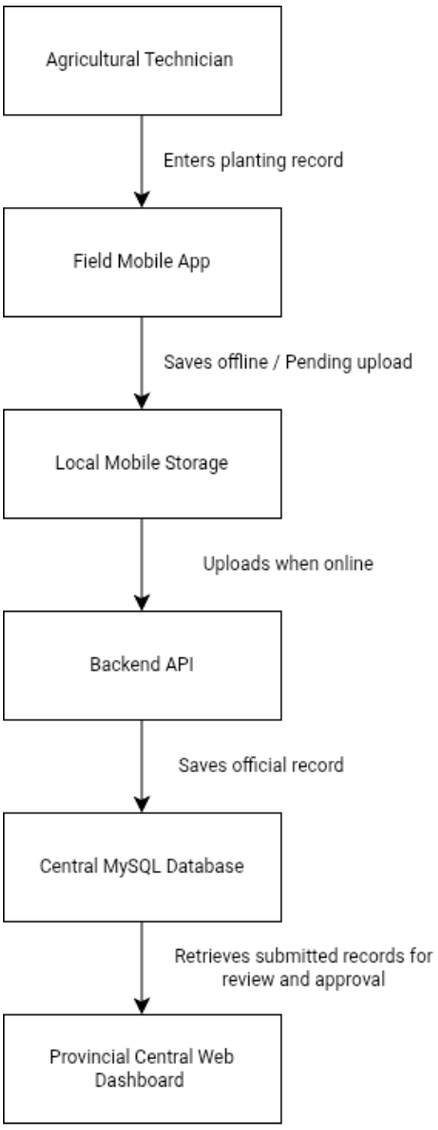
The backend API processes requests from the mobile app and dashboard, validates user access, and exchanges JSON data with the central MySQL database. The database stored user accounts, planting records, approval status, and summary data.

When a field worker entered and saved a planting record in the mobile app, If the device was offline, the record remained in local storage with a pending upload status. Once internet access was available, the record was uploaded to the backend API and saved in the central database.

Authorized personnel can review the record and mark it as approved or returned. Only approved records were included in official summaries, harvest estimates, supply comparisons, and allocation suggestions.

The mobile app and web dashboard communicated with the backend API using JSON. JSON carried record details, synchronization status, and approval status. The mobile app used local storage for pending records, synchronization status, and user session data.

l



.

Figure 1: . Interface architecture and data-handling flow of AgriCast.

2.4 Technology Selection and Comparative Justification

AgriCast used Flutter for the mobile application, HTML, CSS, and JavaScript for the web dashboard, MySQL for the central database, MySQL Workbench for database management, and Leaflet for map display. These technologies were selected based on project scope, accessibility, dependencies, complexity, compatibility, and the team's ability to develop and maintain the system.

Table 3. Technologies and Alternatives

Technology	Alternative Considered	Why chosen for THIS project	Evidence
Flutter	Native Android development using Java or Kotlin	Selected for the Field Mobile App because it supports multiplatform development from a single codebase and provides widgets for building mobile interfaces.	[23]
HTML	A complete frontend framework such as React or Angular	Selected for the structure of the web dashboard, including forms, tables, navigation, and page content.	[27]
CSS	A ready-made styling library such as Bootstrap	Selected to control the layout, appearance, colors, fonts, and screen-size adjustments of the web dashboard.	[28]
JavaScript	A full frontend framework such as React, Vue, or Angular	Selected for interactive dashboard functions, including filters, form feedback, data requests, and dynamic display of records.	[29]
MySQL	Another relational database such as PostgreSQL or SQLite	Selected as the central database because it can store related records in separate tables and support the system's data requirements.	[25]
MySQL Workbench	Command-line tools or phpMyAdmin	Selected for database design, queries, and database inspection because it provides a visual interface for managing the database.	[26]
Leaflet	Another mapping library such as Google Maps API	Proposed for displaying maps, markers, and selectable barangay information because it is lightweight and works with HTML, CSS, and JavaScript.	[30]

The technologies were selected based on scope, accessibility, dependencies, complexity, compatibility, and team skills. The selected technologies matched the required mobile app, web dashboard, database, and map features. They were also widely documented and commonly used, making them easier to learn and maintain. The system required additional packages for offline storage, synchronization, and map data. The selected technologies were simpler and more manageable for the team compared with more complex frameworks. The technologies needed to work together, especially Flutter, the backend API, MySQL, and Leaflet. The team also considered its familiarity with the technologies and its ability to develop and maintain the system.

Flutter alone does not provide a complete offline-saving and synchronization process, so the researchers needed to select and configure the required packages. Leaflet also does not automatically provide official barangay boundaries, road distances, or market information, so these data must come from suitable sources. The backend framework and local-storage package were not yet finalized, and their selection considered security, compatibility, ease of maintenance, and the team's skills.

E. Implementation and Evaluation Procedure

The researchers first prepared the database, user roles, crop benchmarks, and record-status rules. They then developed the mobile and web interfaces and connected them to the backend. Each feature was developed one at a time and tested to check whether it worked correctly before moving to the next feature. After the basic workflow was completed, additional features were added, such as harvest estimates, maps, supply comparisons, reminders, and allocation suggestions.

The researchers conducted unit, functional, and integration testing. Unit testing checked individual functions and calculations. Functional testing checked whether the system features worked according to the intended workflow. Integration testing checked whether the connected system components worked together.

Each feature was tested using its expected behavior. Calculation tests compared system results with separately prepared answers. Integration testing followed a record from mobile entry to uploading, approval, and dashboard display to check whether the different parts worked together correctly.

Offline testing included the following steps:

1. Creating a record without an internet connection to check whether the mobile app can save data locally.
2. Closing and reopening the application to check whether the saved record remained available.
3. Restoring the internet connection to check whether the application can detect connectivity.
4. Uploading the record to check whether the record can be sent to the backend.
5. Checking for missing or duplicate records to confirm that the upload process worked correctly.

The approval process was tested to confirm that unapproved records did not appear in official summaries.

The testing was limited to the features and devices available during development. Some features, such as official barangay boundaries, road distances, and market information, depended on external data sources and were not fully evaluated.

III. RESULTS AND DISCUSSION

3.1 Core Feature 1: Field Data Collection and Offline Recording

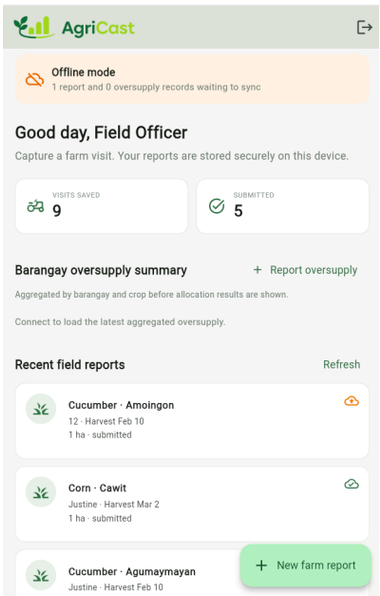
This feature is intended to improve how field personnel record and submit planting information.

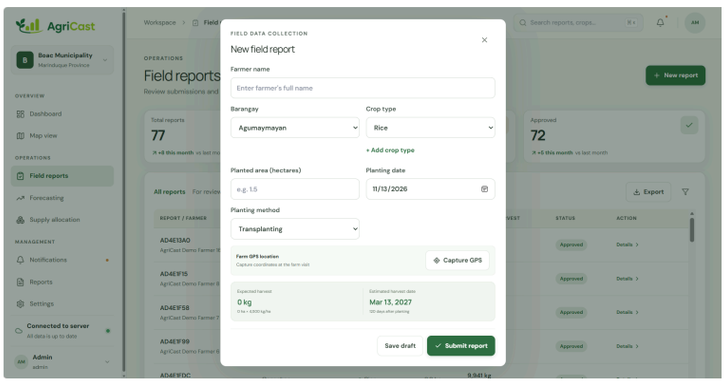
Users will input their name, select the barangay and crop, enter planting details, record the location, and save the record. Required fields and invalid entries will be checked before saving.

The app will be designed to keep records without the internet and upload them when a connection becomes available. Upload status and approval status will be shown separately.

Implementation evidence: Insert the actual planting-form and synchronization screenshots as Figure 1 and Figure 2.

Offline-first guidance supports the proposed approach, but it does not prove that AgriCast has successfully implemented the feature [22].



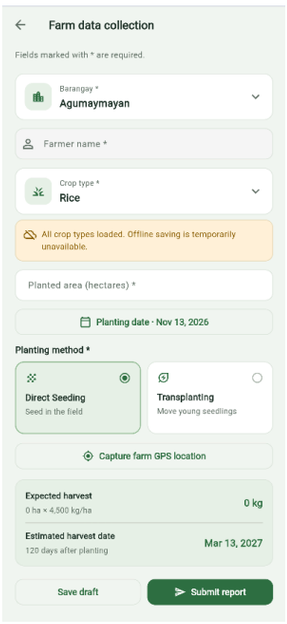


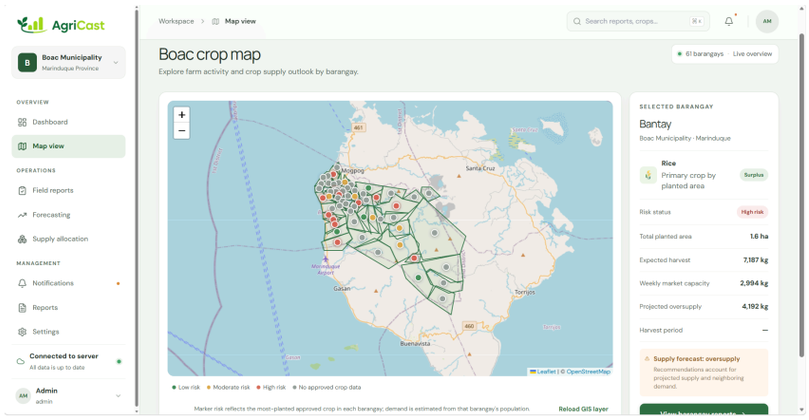
3.2 Core Feature 2: Harvest Estimation and Geographic Monitoring

This feature will help users review expected harvests and crop records by barangay.

Harvest dates and quantities will be estimated using planting records and configured benchmarks. Approved records will be displayed through a map.

Correct calculations do not guarantee accurate harvest predictions. Actual harvests can still differ because the initial method does not include all conditions that affect production [14], [15], [16].



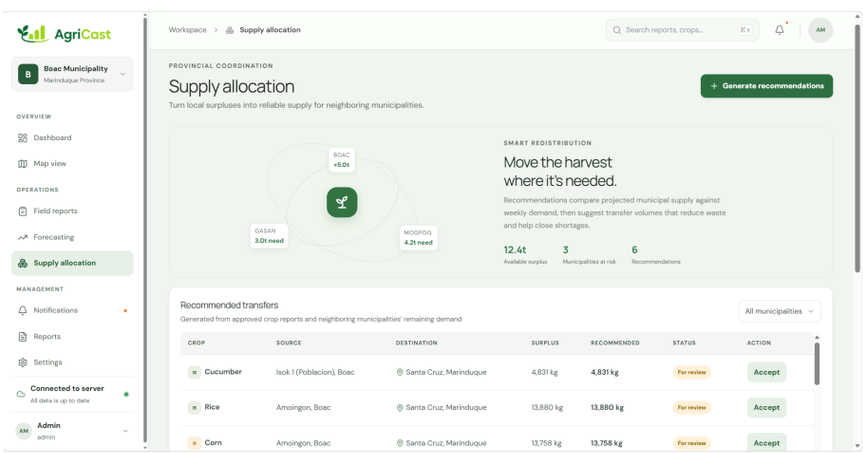


3.3 Core Feature 3: Supply Review and Advisory Allocation

This feature will help users compare expected harvests with estimated consumption and review possible destinations for surplus produce.

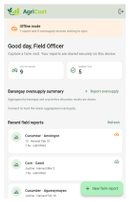
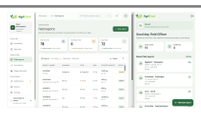
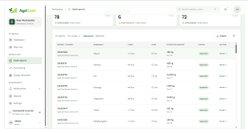
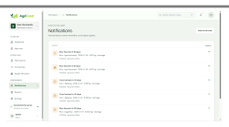
The comparison must use the same crop, location, and period. The system will suggest destinations based on the available information, but officers will make the final coordination decisions.

Agricultural outlook reports support considering production and consumption together, but they do not confirm that a particular local allocation is suitable [17].



3.4 Functional Verification and Limitations

Table 4. Functional Verification Record

Feature	Expected behavior	Available evidence	Status
Planting-record entry	Valid records are saved, and missing required fields are rejected.		Pass
Offline saving	Records remain after restarting the app without internet.		Pass
Synchronization	Records upload without duplication, and failures are shown.		Pass
Approval process	Only approved records appear in official summaries.		Pass
Harvest calculation	Results match independently checked calculations.		Pass
Barangay summaries	Map and table totals match approved records.		Pass
Supply comparison	Crop, location, and period are matched correctly.		Pass
Harvest reminders	Reminders appear within the configured period.		Pass
Allocation suggestions	Suggested quantities do not exceed projected surplus.		Pass
User permissions	Unauthorized actions are blocked.		Pass

IV. CONCLUSION

The preliminary investigation identified a need to improve the collection, checking, updating, and use of crop information in the Office of the Municipal Agriculturist of Boac. Existing recording methods and coordination activities provided a starting point, but the reported problems showed a need to improve the connection between field records, harvest estimates, supply monitoring, and market coordination.

AgriCast was developed to address these needs through a Field Mobile App and a web dashboard. The system was designed to support planting record collection, offline saving, uploading, review and approval, harvest estimation, supply comparison, alerts, and allocation suggestions.

The researchers completed the initial project documents, prepared the system design, selected the main technologies, and developed the main features of the mobile app and web dashboard. The researchers also learned that offline saving, synchronization, approval status, and data consistency required careful planning. The project showed that a simple and organized workflow was needed to connect field records, backend processing, and dashboard summaries.

Although the main features were completed, the system still requires enhancements to improve its performance, usability, and reliability. The formulas for harvest estimates and supply comparisons must be enhanced to make them more accurate. The offline synchronization process, crop benchmarks, and data sources must also be improved using more complete and updated information.

The allocation suggestions may also be improved by considering road distances, receiving capacity, market information, and actual demand. Additional testing and user feedback may help identify further improvements in the system.

REFERENCES — IEEE STYLE

The following references and links are retained from the previous manuscript. Citation numbers are listed individually. Details and links should be checked during the final reference review.

[1] Philippine Statistics Authority, “Cereals statistics of Marinduque 1st semester 2025 and 2026,” Aug. 12, 2026. Available: *https://rssomimaropa.psa.gov.ph/content/cereals-statistics-marinduque-1st-semester-2025-and-2026*

[2] Philippine Statistics Authority, “PSA Marinduque PSO conducts fourth quarter 2025 crops production surveys (households and establishments),” Nov. 20, 2025. Available:*https://rssomimaropa.psa.gov.ph/content/psa-marinduque-pso-conducts-fourth-quarter-2025-crops-production-surveys-households-and*

[3] Philippine Statistics Authority, “2025P fruit crops statistics in Marinduque (Volume of production),” Jun. 5, 2026. Available:*https://rssomimaropa.psa.gov.ph/content/2025p-fruit-crops-statistics-marinduque-volume-production*

[4] Philippine Statistics Authority, “Major vegetables and root crops statistics in MIMAROPA Region fourth quarter 2025,” Jul. 6, 2026. Available:*https://rssomimaropa.psa.gov.ph/content/major-vegetables-and-root-crops-statistics-mimaropa-region-fourth-quarter-2025*

[5] Philippine Statistics Authority, “2025 crops statistics in Marinduque, 1st semester 2024 and 2025,” statistical infographic, n.d. Available:*https://rssomimaropa.psa.gov.ph/system/files/infographics/2025%20CROPS%20STATISTICS.pdf*

[6] Provincial Government of Marinduque, Annual Report 2025, 2026. Available:*https://marinduque.gov.ph/wp-content/uploads/2026/04/Annual-Report-2025-compressed-1.pdf*

[7] R. M. Briones, I. M. R. Galang, and J. S. Latigar, “Transforming Philippine agri-food systems with digital technology: Extent, prospects, and inclusiveness,” PIDS Discussion Paper Series No. 2023-29, 2023. Available:*https://pidswebs.pids.gov.ph/CDN/document/pidsdps2329.pdf*

[8] Food and Agriculture Organization of the United Nations, The State of Food and Agriculture 2022: Leveraging Automation in Agriculture for Transforming Agrifood Systems. Rome: FAO, 2022. Available:*https://www.fao.org/science-technology-and-innovation/resources/publications/the-state-of-food-and-agriculture-2022/*

[9] Food and Agriculture Organization of the United Nations and International Telecommunication Union, Status of Digital Agriculture in 47 Sub-Saharan African Countries. Rome, 2022. Available:https://www.itu.int/dms_pub/itu-d/opb/str/D-STR-DIG_AFR.01-01-2022-PDF-E.pdf

[10] J. Lowenberg-DeBoer, "Economics of adoption for digital automated technologies in agriculture," FAO Agricultural Development Economics Working Paper No. 22-10, 2022. Available:https://www.fao.org/agrifood-economics/publications/detail/ar/c/1613484/9

[11] Food and Agriculture Organization of the United Nations, "Transforming agriculture with digital automation," FAO Agricultural Development Economics Policy Brief No. 54, 2022. Available:https://openknowledge.fao.org/server/api/core/bitstreams/36311f65-6066-4af6-a809-8b285cc1ab07/content

[12] European Commission, Joint Research Centre, "Digitalisation in agriculture: A scoping review of technologies in practice, challenges, and opportunities for smallholder farmers in sub-Saharan Africa," Knowledge4Policy, publication summary, Oct. 15, 2024. Available:https://knowledge-for-policy.ec.europa.eu/publication/digitalisation-agriculture-scoping-review-technologies-practice-challenges_en

[13]Q. Lyu, X. Dai, B. Zhou, F. Wang, L. Dai, H. Xiao, and J.-Q. Ni, "Mapping competences for digital empowerment: A systematic review of smallholder farmers' adoption of digital technologies," Universal Access in the Information Society, vol. 24, pp. 3347-3358, 2025, doi: 10.1007/s10209-025-01238-y. Available:https://doi.org/10.1007/s10209-025-01238-y

[14] R. K. Sharma et al., "Maize and soybean yield prediction using machine learning methods: A systematic literature review," Discover Agriculture, 2025, doi: 10.1007/s44279-025-00215-6. Available:https://doi.org/10.1007/s44279-025-00215-6

[15] S. M. Shawon, F. B. Ema, A. K. Mahi, F. L. Niha, and H. T. Zubair, "Crop yield prediction using machine learning: An extensive and systematic literature review," Smart Agricultural Technology, vol. 10, Art. no. 100718, 2025. Available: https://agris.fao.org/search/en/providers/122436/records/6798b56ed6a63682f044df43

[16] Ajith S, M. K. Debnath, and Karthik R, "Statistical and machine learning models for location-specific crop yield prediction using weather indices," International Journal of Biometeorology, vol. 68, no. 12, pp. 2453-2475, 2024, doi: 10.1007/s00484-024-02763-w. Available: https://doi.org/10.1007/s00484-024-02763-w

[17] Organisation for Economic Co-operation and Development and Food and Agriculture Organization of the United Nations, OECD-FAO Agricultural Outlook 2024-2033. Paris and Rome, 2024. Available:https://doi.org/10.1787/4c5d2cfb-en

[18] Food and Agriculture Organization of the United Nations, The State of Food and Agriculture 2024: Value-Driven Transformation of Agrifood Systems. Rome: FAO, 2024. Available:https://www.fao.org/agrifood-economics/publications/detail/zh/c/1722598/

[19] Food and Agriculture Organization of the United Nations, The State of Food and Agriculture 2025: Addressing Land Degradation Across Landholding Scales. Rome: FAO, 2025. Available:https://www.fao.org/publications/fao-flagship-publications/the-state-of-food-and-agriculture/2025/en

[20] Food and Agriculture Organization of the United Nations, International Fund for Agricultural Development, United Nations Children's Fund, World Food Programme, and World Health Organization, The State of Food Security and Nutrition in the World 2025: Addressing High Food Price Inflation for Food Security and Nutrition. Rome, 2025. Available: https://doi.org/10.4060/cd6008en

[21] Google, "Guide to app architecture," Android Developers. Available:https://developer.android.com/topic/architecture

[22] Google, "Build an offline-first app," Android Developers. Available: https://developer.android.com/topic/architecture/data-layer/offline-first

[23] Flutter, "Multi-platform." Available: https://flutter.dev/development

[24] Flutter, "Testing Flutter apps." Available: https://docs.flutter.dev/testing/overview

[25] Oracle, MySQL 8.4 Reference Manual. Available: https://dev.mysql.com/doc/refman/8.4/en/

[26] Oracle, "MySQL Workbench Manual." Available: https://dev.mysql.com/doc/workbench/en/

[27] MDN Web Docs, "HTML: HyperText Markup Language." Available: https://developer.mozilla.org/en-US/docs/Web/HTM

[28] MDN Web Docs, "CSS: Cascading Style Sheets."Available: https://developer.mozilla.org/en-US/docs/Web/CSS

[29] MDN Web Docs, "What is JavaScript?"Available: https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Scripting/What_is_JavaScript

[30] Leaflet, "API reference."Available: https://leafletjs.com/reference.html

CRediT AUTHOR CONTRIBUTIONS STATEMENT

Project: AgriCast: Crop Supply Forecasting and Allocation System

Group Name: Nimloi

Course / Section: BSI/T 3C

Instructions: Mark the relevant roles for all 5–6 members. Multiple people may share a role and any contributor may have multiple roles. Each member should verify their entries. CRediT documents contributions; it does not determine who qualifies for authorship.

Contributor	C	M	So	Va	Fo	I	R	D	O	E	Vi	Su	P	Fu
Author 1: Romasanta, Sheila Mae M.	☑	☑	☐	☑	☑	☑	☑	☐	☑	☑	☐	☑	☑	☑
Author 2: Laylay, John Lloyd Q.	☑	☑	☐	☐	☐	☑	☑	☐	☑	☑	☑	☐	☐	☑
Author 3: Malco, John Cedrick R.	☐	☑	☑	☑	☐	☑	☑	☑	☐	☐	☐	☐	☑	☑
Author 4: Zulueta, Aron M.	☐	☑	☑	☐	☐	☑	☑	☐	☑	☑	☑	☐	☐	☑
Author 5: Medallon, Benedict O	☐	☑	☑	☐	☐	☑	☑	☐	☑	☑	☑	☐	☐	☑
Author 6: Montellano, Ken Gabriel LB.	☐	☑	☐	☐	☐	☑	☑	☐	☐	☑	☐	☐	☐	☑

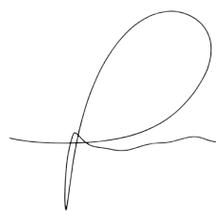
ROLE LEGEND

Code	Role	Code	Role
C	Conceptualization	M	Methodology
So	Software	Va	Validation
Fo	Formal analysis	I	Investigation

R	Resources	D	Data curation
O	Writing – original draft	E	Writing – review & editing
Vi	Visualization	Su	Supervision
P	Project administration	Fu	Funding acquisition

Member confirmation:

We, the members of the group, confirm our participation and acknowledge our respective responsibilities and contributions to the project. By signing below, we agree to the decisions and tasks assigned to us.

Name of Member	Date Confirmed	Signature
ROMASANTA, Sheila Mae M.	November 13, 2026	
LAYLAY, John Lloyd Q.	November 13, 2026	
MALCO, John Cedrick R.	November 13, 2026	
ZULUETA, Aron M.	November 13, 2026	
MEDALLON, Benedict O.	November 13, 2026	
MONTELLANO, Ken Gabriel LB.	November 13, 2026	

Credit role definitions: https://credit.niso.org/contributor-roles/ . Roles marked must reflect substantive actual contributions; no role is mandatory by default.

AI USE AND SUPPORTING EVIDENCE

AI tools were used in preparing this manuscript. AI was used only to assist in organizing and revising some parts of the text, such as improving clarity and formatting. The researchers reviewed and revised all AI-assisted outputs before including them in the manuscript.

AI tools were also used in the development of the AgriCast system. These tools assisted in generating code suggestions, debugging errors, and explaining technical functions during development. The researchers reviewed, tested, and revised the generated outputs before using them in the system.